

$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \Rightarrow k=4$

first = $A[0] = 1$

last = $A[8] = 9$

middle value = $A[(0+8)/2] = A[4] = 5$

Median of the three = $\{1, 9, 5\}$ Pivot = 5

1 2 3 4 5 6 7 8 9 (5)

i → i → i → i → i → i → i → i → i
1 2 3 4 5 6 7 8 9 (5)

1 2 3 4 | 5 | 6 7 8 9

L=4 E=1 G=4

Since $|L| + |E| = 5 > k=4$, k remains 4

Recursively call QuickSort(L, 4)

1 2 3 4

k=4

first = $A[0] = 1$

last = $A[3] = 4$

middle value $A[(0+3)/2] = A[1] = 2$

Median of the three $\{1, 4, 2\} \Rightarrow$ Pivot = 2

1 2 3 4 (2)

1 2 3 4 (2)

1 | 2 | 3 4

L=1 E=1 G=2

Since $|L| + |E| = 2 < k=4$, the k value changes to

$k - (|L| + |E|) = 4 - (1+1) = k=2$

Recursively Call QuickSelect(G, 2)

3 4

k=2

first = $A[0] = 3$

last = $A[1] = 4$

middle value = $A[(0+1)/2] = A[0] = 3$

Median of the three = $\{3, 4, 3\} \Rightarrow$ Pivot = 3

3 4 (3)

3 4 (3)

3 | 4

L=0 E=1 G=1

Since $|L| + |E| = 1 < k=2$, the k value changes to

$k - (|L| + |E|) = 2 - (0+1) = k=1$

Recursively call QuickSelect
 $A = \{4\}$

It returns 4, since
has only one element

$A = \{8, 7, 6, 5, 4, 3, 2, 1, 9\}, k = 5$

first $A[0] = 8$

last $A[8] = 9$

middle value $= A[(0+8)/2] = A[4] = 4$

Median of the three $\{8, 9, 4\} \Rightarrow \underline{\text{Pivot} = 8}$

9 7 6 5 4 3 2 1 (8)

9 7 6 5 4 3 2 1 (8)

1 7 6 5 4 3 2 9 (8)

[1 7 6 5 4 3 2] (8) [9]

$L = 7, E = 1, G = 1$

Since $|L| + |E| = 8 > k = 5$, k remains 5

Reversibly call QuickSelect (2, 5)

1 7 6 5 4 3 2

$k = 5$

first $A[0] = 1$

last $A[6] = 2$

middle value $A[(0+6)/2] = A[3] = 5$

Median of the three $\{1, 2, 5\} \Rightarrow \underline{\text{Pivot} = 2}$

1 7 6 5 4 3 (2)

1 7 6 5 4 3 (2)

[1] (2) [6 5 4 3 7]

$L = 1, E = 1, G = 5$

Since $|L| + |E| = 1 + 1 = 2 < k = 5$, the k value changes to

$k - (|L| + |E|) = 5 - (1 + 1) = \underline{k = 3}$

Reversibly call QuickSort (6, 3)

6 5 4 3 7

$k = 3$

first $A[0] = 6$

last $A[4] = 7$

middle value $A[(0+4)/2] = A[2] = 4$

Median of the three $\{6, 7, 4\} \Rightarrow \underline{\text{Pivot} = 6}$

7 5 4 3 (6)

7 5 4 3 (6)

3 5 4 7 (6)

[3 5 4] (6) [7]

$L = 3, E = 1, G = 1$

Since $|L| + |E| = 4 > k = 3$, k remains 3

Reversibly call QuickSort (2, 3)

3 5 4

$k = 3$

first $A[0] = 3$

last $A[2] = 4$

middle value $A[(0+2)/2] = A[1] = 5$

Median of the three $\{3, 4, 5\} \Rightarrow \underline{\text{Pivot} = 4}$

3 5 (4)

3 5 (4)

[3] (4) [5]

$L = 1, E = 1, G = 1$

Since $|L| + |E| = 2 < k = 3$, the k value changes to

$k - (|L| + |E|) = 3 - (1 + 1) = \underline{k = 1}$

Reversibly call QuickSelect (6, 1)

$\Rightarrow \{5\}$

I returns 5, since 6 has only one element

$A = \{9, 1, 8, 2, 7, 3, 6, 4, 5\}$, $k = 6$

first $A[0] = 9$

last $A[8] = 5$

middle value $A[(0+8)/2] = A[4] = 7$

Median of the three $\{9, 5, 7\} \Rightarrow \text{Pivot} = 7$

9 1 8 2 5 3 6 4 7

4 1 8 2 5 3 6 7 9

4 1 6 2 5 3 7 9 8

4 1 6 2 5 3 7 9 8

4 1 6 2 5 3 7 9 8

$L = 6$, $E = 1$, $G = 2$

Since $|L| + |E| = 7 > k = 6$, k remains 6

Recursively call QuickSelect($L, 6$)

$A = \{4, 1, 6, 2, 5, 3\}$

$k = 6$

First $A[0] = 4$

last $A[5] = 3$

middle value $A[(0+5)/2] = A[2] = 6$

Median of the three $\{4, 3, 6\} \Rightarrow \text{Pivot} = 4$

3 1 6 2 5 4

3 1 6 2 5 4

3 1 2 6 5 4

3 1 2 4 5 6

$L = 3$, $E = 1$, $G = 2$

Since $|L| + |E| = 4 < k = 6$, the k value changes

$k - (|L| + |E|) = 6 - (3 + 1) = k = 2$

Recursively call QuickSelect($G, 2$)

5 6

$k = 2$

first $A[0] = 5$

last $A[1] = 6$

middle value $A[(0+1)/2] = A[0] = 5$

Median of the three $\{5, 6, 5\} \Rightarrow \text{Pivot} = 5$

6 5

6 5

5 6

$L = 0$, $E = 1$, $G = 1$

Since $|L| + |E| = 1 < k = 2$, the k value changes

$k - (|L| + |E|) = 2 - (0 + 1) = k = 1$

Recursively call QuickSelect($G, 1$)

It return 6, since 6 has only one element.